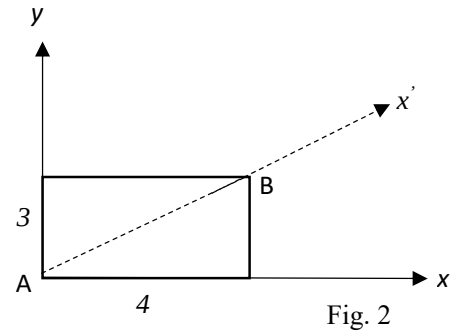
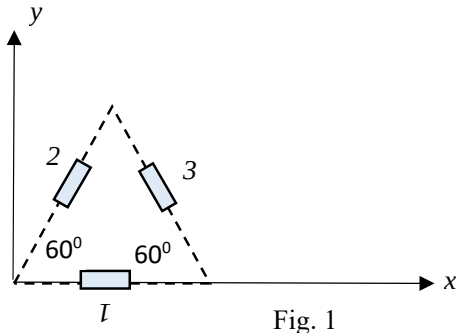


Problems on strain

Problem 1. Three strain gauges are attached to a thin stressed metallic plate as shown in the figure
1. The strain readings are as follows.

$$\varepsilon_1 = 4 \times 10^{-4} ; \varepsilon_2 = 2 \times 10^{-4} \text{ and } \varepsilon_3 = 3 \times 10^{-4}$$

Compute ε_{xx} and ε_{xy} (note $\varepsilon_{xy} = \gamma_{xy}/2$)



Problem. 2 A thin rectangular palte shown in figure 2 of dimension 4 x 3 units is subjected to uniform strains given by,

$$\varepsilon_x = 5 \times 10^{-3} ; \varepsilon_y = 2.5 \times 10^{-3} \text{ and } \gamma_{xy} = 1 \times 10^{-3}$$

Determine change in length of the diagonal AB.

Problem 3. In Cartesian co-ordinate system, the displacement vector (u, v, w) is given as

$$U(x, y, z) = \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} x \\ y + 3z \\ 8x + 3z \end{bmatrix} 10^{-3}$$

Determine a) strain tensor at $(x=1, y=1 \text{ and } z=1)$; b) Strain invariants and c) principal strains and verify the strain invariants using principal strains.

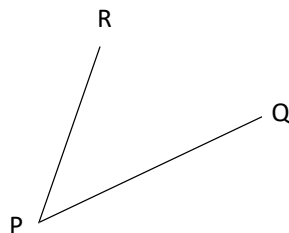
Problem 4. The displacement field in a 3d body is given by,

$$u = (x^2 + 2z) \times 10^{-3} ; v = (3x + 2y + 2z) \times 10^{-3}; \text{ and } w = (3z^2) \times 10^{-3}$$

Compute change in the angle between two line segments PQ and PR at $P(2,1,3)$ having direction cosine before deformation as

$$\text{PQ: } l_1 = 0; m_1 = n_1 = \frac{1}{\sqrt{2}}$$

$$\text{PR: } l_2 = 0.6; m_2 = 0; n_2 = 0.8$$



Problem 5: The displacement field in a 3d body is given by

$$\mathbf{u} = (x^2 + 2z) \times 10^{-3} ; \mathbf{v} = (3x + 2y) \times 10^{-3}; \text{ and } \mathbf{w} = (3z) \times 10^{-3}$$

Compute normal strain at (1,2,3) in the direction
 $\mathbf{t} = 0.60 \mathbf{i} + 0.8 \mathbf{k}$